

Data Analytics Mini Project

Gold Price Trend and Market Analysis using Excel and Power BI

1. Project Objective

To perform data cleaning and transformation in Excel and use the prepared dataset to design an interactive Power BI dashboard that communicates gold-price trends, volatility, trading volume and the relationship between gold prices and the USD Index.

2. Dataset Selection

Dataset: Gold Price 1970–2026 Daily. The supplied cleaned workbook contains 20,512 records and 17 features. The original dataset contains 7 source fields: Date, Open, High, Low, Close, Volume and USD_Index.

3. Excel Data Pre-processing

- Verified the supplied Data Quality sheet. It reports 20,819 original rows, 307 completely blank rows removed and 0 duplicate rows removed.

	A	B
1	Data Quality Check	Result
2	Original rows	20819
3	Completely blank rows removed	307
4	Duplicate rows removed	0
5	Final valid records	20512
6	Missing values in final dataset	0
7	Date range	01-01-1970 to 27-02-2026
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Cleaned Data

Data Quality

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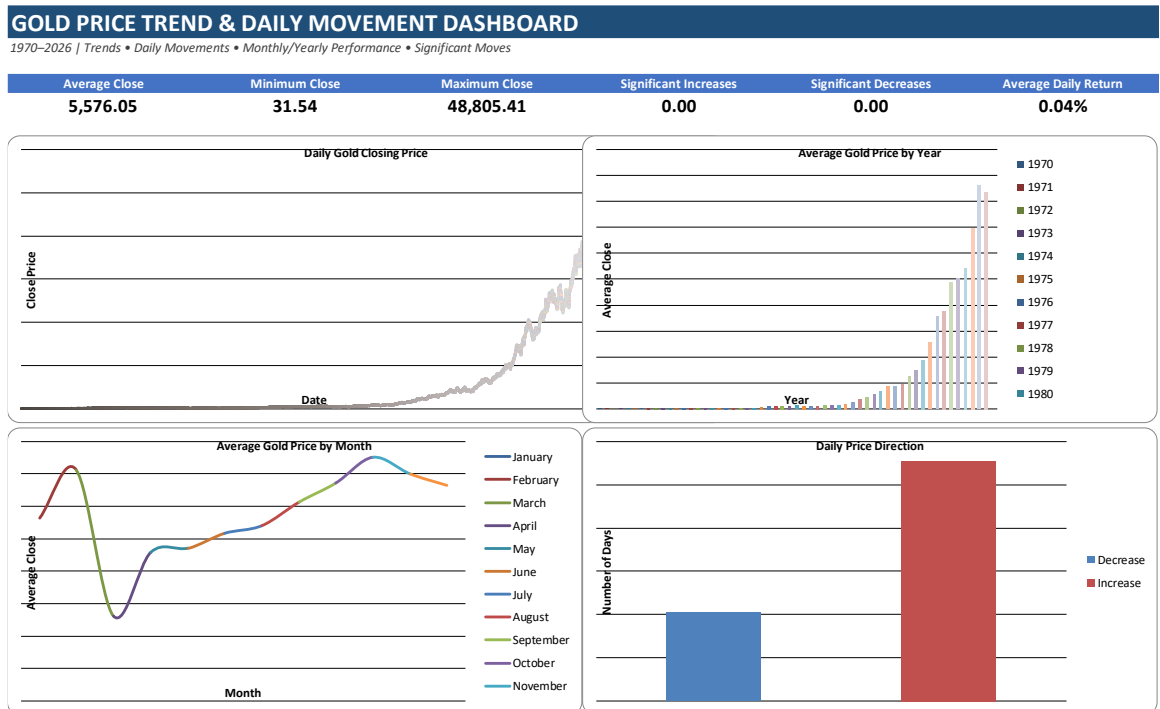
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- Retained the core market fields and added Year, Month, Month Name, Quarter, Day, Day of Week and Year-Month for time-based analysis.

Date	Open	High	Low	Close	Volume	USD Index	Year	Month	Month Name	Quarter	Day	Day of Week	Year-Month	Price Change	Change %	Price Range
1970-01-01 00:00:00	35.26647996	35.34704772	35.02873204	35.18483631	463815	101.65397776	1970	1	January	Q1	1	Thursday	1970-01	-0.08164365	-0.231504959	0.31831568
1970-01-02 00:00:00	35.11844001	35.21059766	35.11303282	35.1467643	118078	99.60223676	1970	1	January	Q1	2	Friday	1970-01	0.02832429	0.080653611	0.09756484
1970-01-03 00:00:00	35.39048113	35.40159509	35.30879502	35.3857598	268897	99.37285493	1970	1	January	Q1	3	Saturday	1970-01	-0.00472233	-0.013345503	0.09380007
1970-01-04 00:00:00	35.87715992	36.12963408	35.81995377	35.93959972	441570	99.68925768	1970	1	January	Q1	4	Sunday	1970-01	0.0624398	0.174037745	0.30968031
1970-01-05 00:00:00	35.8497571	35.91334706	35.72434475	35.86630266	443895	99.94306702	1970	1	January	Q1	5	Monday	1970-01	0.01654556	0.046152502	0.18000231
1970-01-06 00:00:00	35.79758727	35.98335499	35.6982166	35.79316096	369914	100.0178396	1970	1	January	Q1	6	Tuesday	1970-01	-0.00442631	-0.012364828	0.28513839
1970-01-07 00:00:00	36.36763169	36.58137992	36.33175013	36.37380849	367899	100.1354445	1970	1	January	Q1	7	Wednesday	1970-01	0.0061768	0.016984334	0.24962979
1970-01-08 00:00:00	36.76345298	36.87496859	36.66420726	36.66502547	339524	101.3316945	1970	1	January	Q1	8	Thursday	1970-01	-0.09842751	-0.267731951	0.21076133
1970-01-09 00:00:00	36.48014055	36.57280516	36.39704623	36.50424562	215043	98.83367926	1970	1	January	Q1	9	Friday	1970-01	0.02410507	0.06607724	0.17575893
1970-01-10 00:00:00	36.64541287	36.75124088	36.64165971	36.71385384	323236	100.8700285	1970	1	January	Q1	10	Saturday	1970-01	0.06844097	0.186755449	0.10958117
1970-01-11 00:00:00	36.49149645	36.59258939	36.44399743	36.55507384	408203	100.6842628	1970	1	January	Q1	11	Sunday	1970-01	0.06357739	0.17422522	0.14859196
1970-01-12 00:00:00	36.44382526	36.58162428	36.34109813	36.39613902	243082	100.3038523	1970	1	January	Q1	12	Monday	1970-01	-0.04768624	-0.130848613	0.24052615
1970-01-13 00:00:00	36.39284171	36.51990746	36.33576378	36.49525751	364828	100.4336346	1970	1	January	Q1	13	Tuesday	1970-01	0.1024158	0.28141743	0.18414368
1970-01-14 00:00:00	35.79997035	35.91147717	35.65181597	35.81438106	136317	99.53717524	1970	1	January	Q1	14	Wednesday	1970-01	0.01441071	0.040253413	0.2596612
1970-01-15 00:00:00	35.3058612	35.33369911	35.15901447	35.21247207	135907	98.59322755	1970	1	January	Q1	15	Thursday	1970-01	-0.09348913	-0.264797011	0.17468464
1970-01-16 00:00:00	35.04603527	35.14312699	35.00347668	35.02533843	402696	100.3679009	1970	1	January	Q1	16	Friday	1970-01	-0.02049684	-0.058485474	0.13965031
1970-01-17 00:00:00	34.67780759	34.79198697	34.52528181	34.68298267	115418	99.5096486	1970	1	January	Q1	17	Saturday	1970-01	0.00517508	0.014923319	0.26670516
1970-01-18 00:00:00	34.8000841	34.87827016	34.75773673	34.80258365	497521	100.5664894	1970	1	January	Q1	18	Sunday	1970-01	0.00249955	0.007182598	0.12053343
1970-01-19 00:00:00	34.42730963	34.51901914	34.39885532	34.49834618	179655	99.66608806	1970	1	January	Q1	19	Monday	1970-01	0.07103655	0.20633779	0.12016382
1970-01-20 00:00:00	33.96642745	34.09455877	33.90677394	34.02475504	273048	100.8399809	1970	1	January	Q1	20	Tuesday	1970-01	0.05832759	0.171721298	0.18778483
1970-01-21 00:00:00	34.52398836	34.56256802	34.51157182	34.53747053	263557	101.0464802	1970	1	January	Q1	21	Wednesday	1970-01	0.01348217	0.0390516	0.0509962

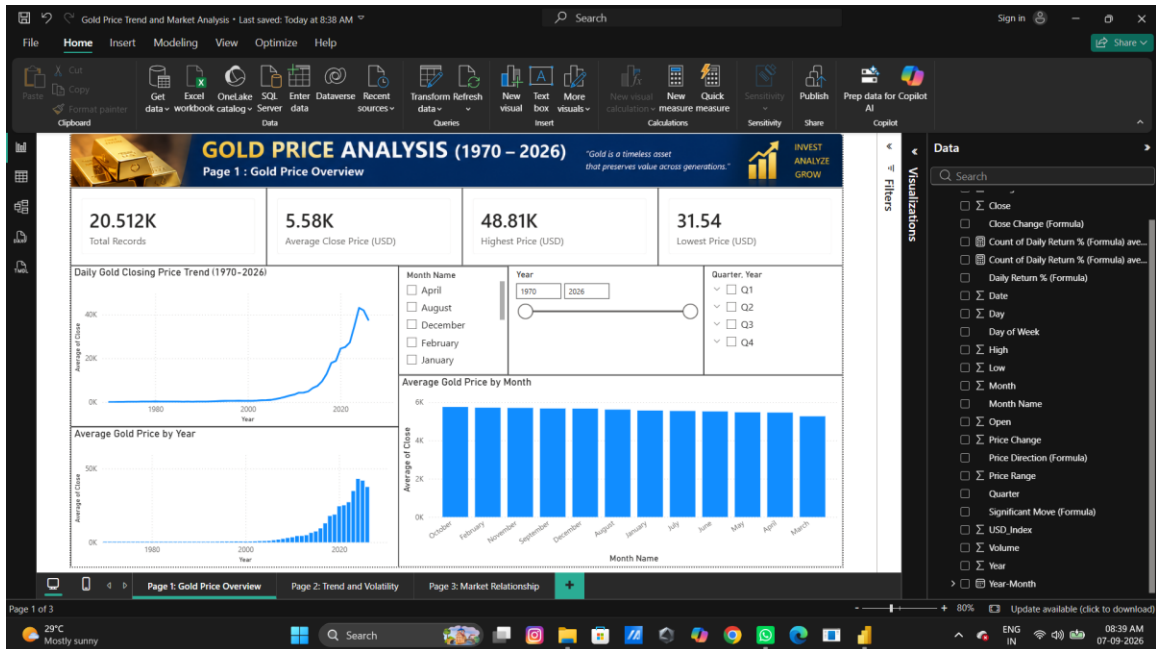
- Created Price Change, Change % and Price Range to support movement and volatility analysis.
- Formatted Date as a date field and numerical market fields as numeric values suitable for Power BI.
- Applied filtering/table structure and conditional formatting to improve usability.



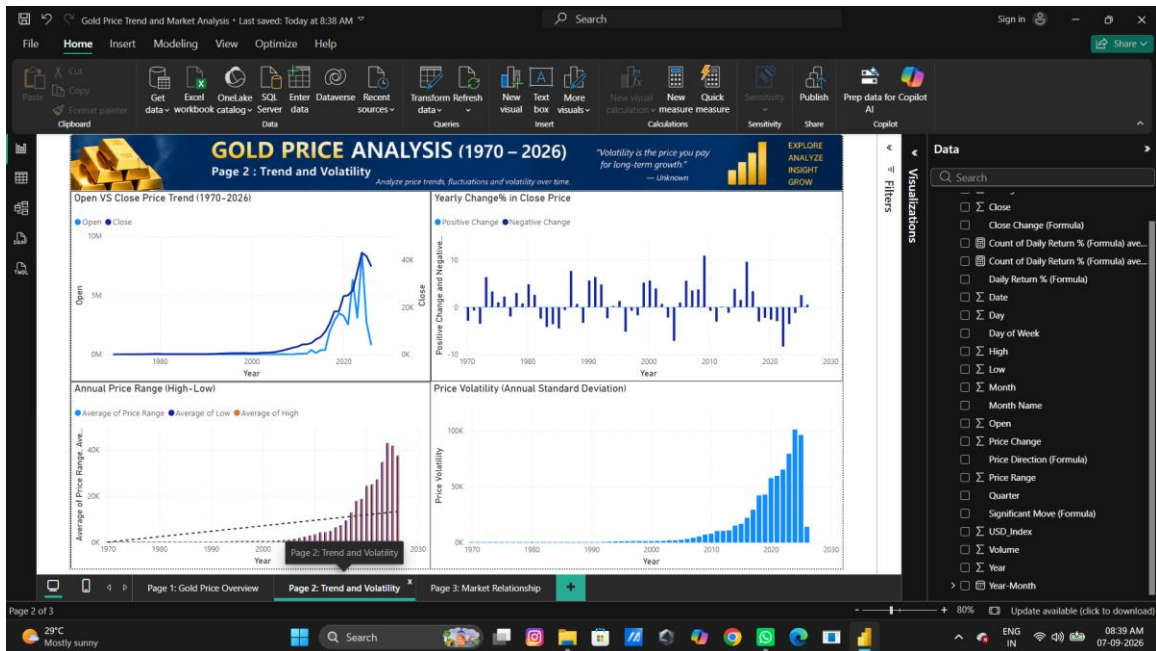
Dashboard use:
Use the Pivot Tables sheet for Excel pivot-style summaries and the Dashboard sheet for the presentation view.

4. Power BI Dashboard Design

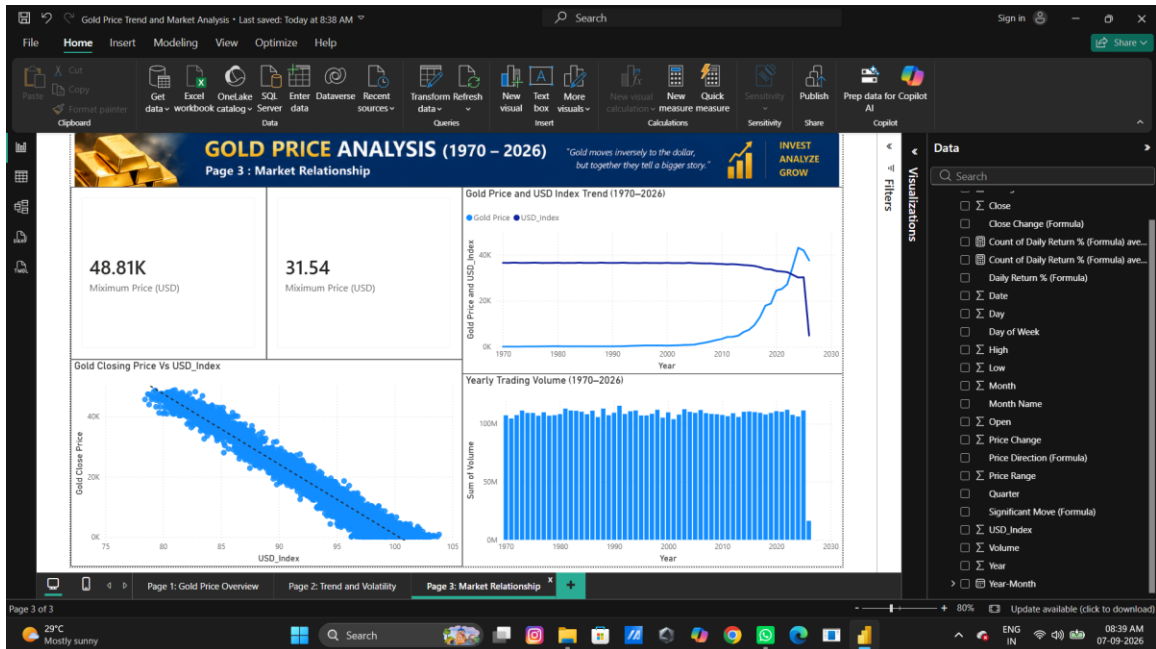
Page 1: Gold Price Overview. Use KPI cards, a daily close-price line chart, an annual average-price column chart, and Year/Quarter/Month slicers.



Page 2: Trend and Volatility. Compare Open vs Close, annual Price Range, and Change % using line and column charts.



Page 3: Market Relationship. Use a scatter chart for Close versus USD Index, a yearly volume chart, and KPI cards for maximum and minimum price dates.



5. Key Findings from the Supplied Cleaned Dataset

- Highest recorded Close: 48,805.41 on 09-06-2024.
- Lowest recorded Close: 31.54 on 22-02-1970.
- Highest average annual Close occurred in 2024.
- Close price and USD Index show a strong negative correlation of -0.975 in this dataset.
- Cleaned dataset contains no blank rows or duplicate rows according to the supplied Data Quality sheet.

6. Conclusion

The supplied dataset has been structured for downstream Power BI analysis. The workbook contains the cleaned data, data-quality information, project summary, dashboard design specification and DAX measures. The Power BI PDF submission should be exported after implementing these visuals in Power BI Desktop.

Source workbook used: Gold_Price_Cleaned.xlsx